

RecoMart End-to-End Data Management Pipeline

This report documents the complete implementation for Assignment I, including ingestion, validation, preparation, transformation, feature management, data versioning, model training, and orchestration.

Rubric Area	Implementation Evidence
Problem Formulation	Problem report, objectives, sources, outputs, metrics
Pipeline Implementation	CSV and REST ingestion, raw partitions, validation, preparation, transformations
Feature Store & Versioning	Custom versioned feature store, metadata registry, DVC-compatible workflow
Model Training & Evaluation	SVD recommender, Precision@10, Recall@10, NDCG@10, run metadata
Documentation & Demo	README, final report, logs, demo script

Data Quality Summary

Dataset	Rows	Columns	Issues
interactions	3507	8	3
products	120	6	0

Feature Tables

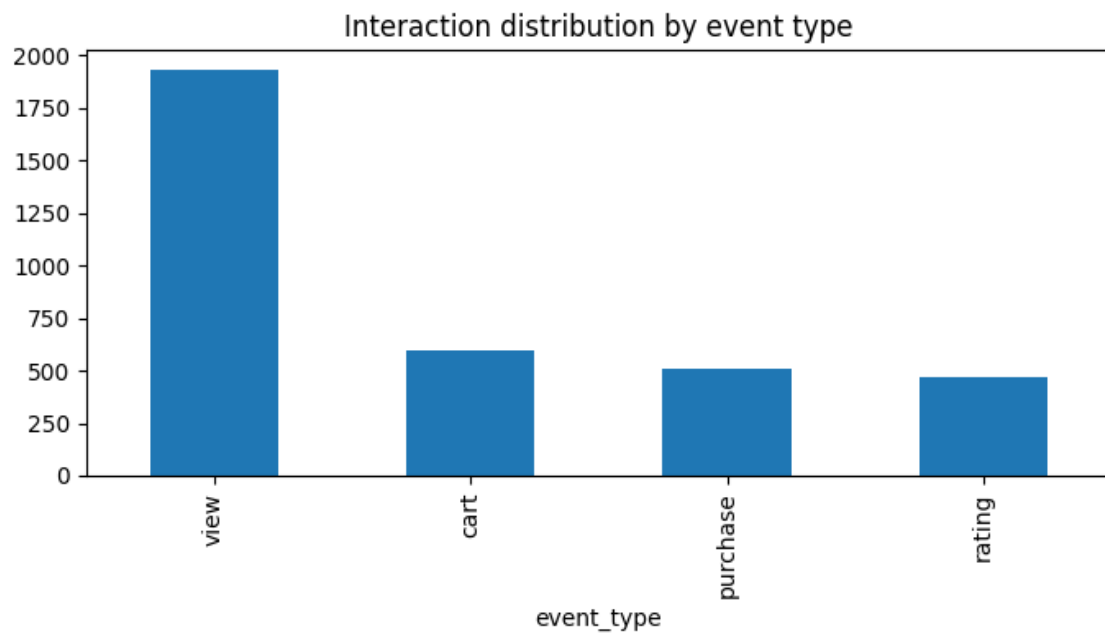
Table	Rows
user_features	240
item_features	120
user_item_features	2946
item_cooccurrence	600

Model Performance

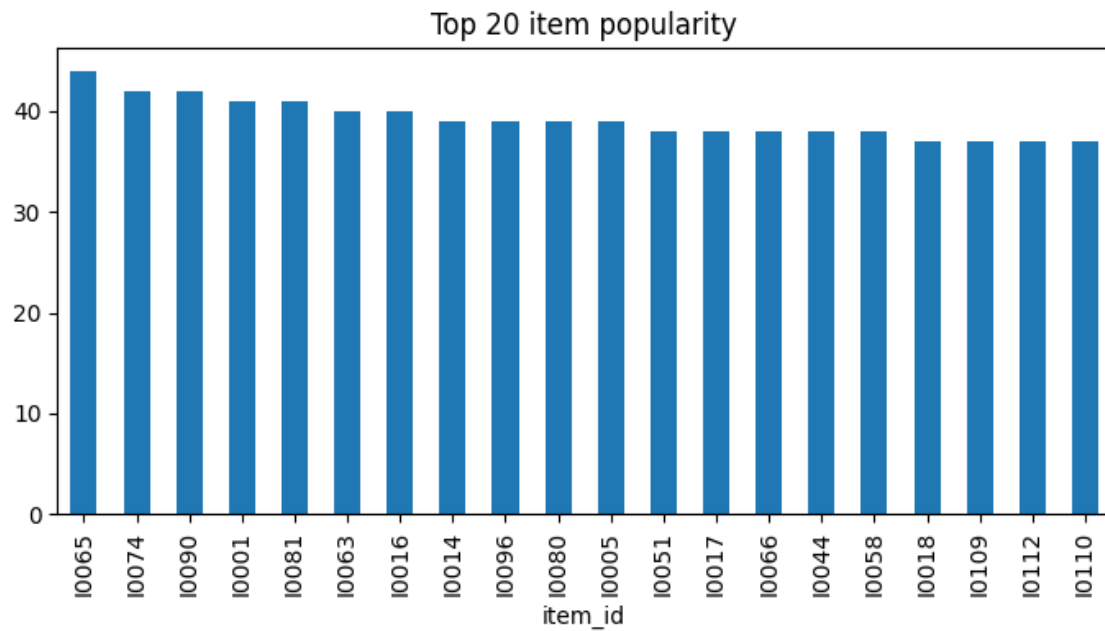
Metric	Value
precision_at_10	0.0315
recall_at_10	0.2366
ndcg_at_10	0.1075
evaluated_users	124
train_rows	3114
test_rows	385

EDA Plots

interaction_distribution.png



item_popularity_top20.png



sparsity_heatmap.png

User-item matrix sample heatmap

